Lab Exercise 1: The Dipole Antenna

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Exercise 1.1: Analytical Design

At $f = 1$ GHz, the free-space wavelength is

$$\lambda = \frac{c}{f} = \frac{2.998 \times 10^8 \text{ m/s}}{1 \times 10^9 \text{ Hz}} = 299.8 \text{ mm}$$

so an exact half wave is $\lambda/2 = 149.9$ mm.

Table 1: Initial design parameters.

Parameter	**Symbol**	**Value**	**Reason**
Total length (tip to tip)	L	149.9 mm	A half wave dipole is simple and effective, so we shall use it in this lab.
Conductor	—		*[...]*
Wire radius (or strip width)	a (or w)	*[...]* mm	*[...]*
Feed gap	g	*[...]* mm	*[match adapter terminal spacing]*

> **Question 1.** What initial dimensions would you select for a dipole resonant near 1 GHz and why?

Answer. *[Answer here.]*

> **Question 2.** What input resistance do you expect at resonance, and how does this compare with the 50 Ω impedance of the VNA?

Answer. *[Answer here.]*

> **Question 3.** How should the antenna impedance change as the length is increased or decreased?

Answer. *[Answer here.]*

> **Question 4.** How might the conductor diameter or width affect resonance and bandwidth?

Answer. *[Answer here.]*

Exercise 1.2: CST Simulation

Project setup

1. **New project.** `File → New and Recent → New Template → Microwaves & RF/Optical → Antennas → Wire → Time Domain`. Units: mm, GHz, ns. Frequency range 0.7 to 1.3 GHz (a narrow band keeps the mesh small, since CST sizes cells by the wavelength at $f_{\max}$).
2. **Field monitors** (the template wizard asks; or add later from `Simulation → Field Monitor`):
   - H-field / surface current at 0.9, 1.0, 1.1 GHz (surface current is derived from the H-field monitor).
   - Farfield at 1.0 GHz (for radiation efficiency later).
3. **Parameters.** In the `Parameter List` at the bottom, add:
   - `L` = total length (from Table 1)
   - `a` = wire radius
   - `g` = feed gap

   Build everything below from these names so the sweeps update the geometry.
4. **Boundaries.** `Simulation → Boundaries`: all six faces **open (add space)**. The template already does this; confirm it, and check that the background adds roughly $\lambda/4$ of space around the antenna.

Geometry (round wire)

1. **Upper arm.** `Modeling → Shapes → Cylinder`, press Esc to type values:
   - Orientation: Z. Outer radius `a`, inner radius 0.
   - Xcenter = Ycenter = 0. Zmin = `g/2`, Zmax = `L/2`.
   - Segments: 0 for a true circle. If you hit the 2000-element limit, set 8 (octagonal prism), as in Exercise 4.
   - Material: PEC to start.
2. **Lower arm.** Same dialog with Zmin = `-L/2`, Zmax = `-g/2`. Name it as a separate solid so the two arms are not merged.

Copper-tape alternative: `Shapes → Brick` with X from `-w/2` to `w/2`, Y from 0 to `t` (tape thickness, about 0.04–0.07 mm; measure yours), Z from `g/2` to `L/2`, and mirrored for the other arm. Add a foam board brick behind it (`Materials → New`: $\varepsilon_r \approx 1.05$–$1.1$). Thin tape is mesh-expensive; if the cell count explodes, model it as a zero-thickness PEC sheet instead.

Feed

1. **Discrete port by coordinates.** `Simulation → Sources and Loads → Discrete Port`. Uncheck **Use pick** for both points:
   - Point 1: (0, 0, `g/2`). Point 2: (0, 0, `-g/2`).
   - Type: S-Parameter. Impedance: 50 Ω. Radius: 0.

   This bridges the two arm end faces along the axis and tracks `g` in sweeps.

2. **Realistic feed (do this after the ideal case works).** Set `g` to the actual spacing of the SMA screw terminals, and add the short leads from the terminals to the arms as extra wire segments. Record which version produced each plot. The coax itself sits behind the calibration plane and is not modeled.

Mesh

1. `Simulation → Mesh → Mesh View`. Check the cell count in the status bar against the license limit.
2. If over the limit, in order:
   - Segments = 8 on both cylinders.
   - `Global Properties`: lower cells per wavelength; raise the minimum cell size.
   - Narrow the band to 0.8–1.2 GHz.
   - Symmetry planes (`Simulation → Boundaries → Symmetry Planes`): YZ and XZ as **magnetic** ($H_t = 0$), since (H) circles the z-directed current and is normal to any plane containing the wire. Drop this if CST complains about the port on the plane.
3. **Convergence check.** `Time Domain Solver → Adaptive mesh refinement` on, or run twice with denser mesh and confirm the zero-reactance frequency moves by less than about 1%. Report this.

Solve and read results

1. `Simulation → Setup Solver`: Time Domain, accuracy −40 dB. Start. If the solver warns that energy has not decayed, increase the number of pulses / simulation duration.
2. **S_{11}.** `1D Results → S-Parameters → S1,1`. Plot in dB and also as Re/Im (right-click → Plot Properties, or the `1D Plot` ribbon). Use curve markers for the $|S_{11}|$ minimum and the −10 dB crossings.
3. **Input impedance.** `1D Results → Z Parameters → Z1,1`, Re and Im. Marker where $\mathrm{Im}(Z_{11})$ crosses zero: that is the zero-reactance frequency; read R there. Cross-check one point by hand with $Z_{\text{in}} = 50(1 + S_{11})/(1 - S_{11})$ using full-precision marker values.
4. **Surface current.** `2D/3D Results → Surface Current → h-field (f=1)`. Show as arrows or abs, with a linear scale. For a distribution along the arm: `Modeling → Curves → Line` from (`a`, 0, `g/2`) to (`a`, 0, `L/2`), then `Post-Processing → Template Based Post-Processing → 1D → Evaluate Field on Curve` on the H-field monitor (tangential H at the surface equals the surface current density).
5. **Efficiency (for Q12 later).** `Farfields → farfield (f=1)` reports Rad. efficiency. It is 100% for PEC. Change the arms to **Copper (annealed)**, rerun, and note the change in R and efficiency.

Parameter sweeps

1. **Length.** `Simulation → Par. Sweep → New Seq. → New Par.`: `L` from 130 to 160 mm in 5 mm steps (finer near resonance once you see it). Add result templates for S1,1 and Z1,1 if your version does not store them automatically. Overlay results with the `Result Navigator`.
2. **Thickness (if time).** Sweep `a` over the radii of the wire gauges available, holding `L` fixed. Look at the zero-reactance frequency and the width of the −10 dB band.

3. **Export.** `File → Export → Touchstone` (or right-click a 1D result → Export → ASCII) for each case, so CST and LibreVNA data can be overlaid in Python with identical axes.

> **Question 5.** How closely does the simulated resonant length agree with your analytical estimate?

Answer. *[Answer here.]*

> **Question 6.** What happens to the input impedance as the dipole length changes?

Answer. *[Answer here.]*

> **Question 7.** How does the predicted bandwidth depend on conductor thickness or width?

Answer. *[Answer here.]*

> **Question 8.** What current distribution is observed along the antenna near resonance?

Answer. *[Answer here.]*

Selected fabrication geometry: *[…]*

Exercise 1.3: Fabrication

Table 2: Design vs. as-built dimensions.

Dimension	Design	As built
Total length L	*[…]*	*[…]*
Radius a / width w	*[…]*	*[…]*
Feed gap g	*[…]*	*[…]*
Lead length	*[…]*	*[…]*

CST update: if any as-built dimension differs appreciably, change the parameters in the `Parameter List` to the measured values and rerun the single-point simulation (no sweep). Use this as the CST curve in the measurement overlays.

Exercise 1.4: Calibration and Measurement

[Calibration range, standards, reference plane, adapter treatment, cable-movement diagnostic.]

Exercise 1.5: Geometry Study

[Lengths measured, CST vs. measured shift in resonance and impedance.]

Exercise 1.6: Comparison with Theory

> **Question 9.** How does the analytical radiation resistance compare with the simulated input resistance?

Answer. *[Answer here.]*

> **Question 10.** How closely does the measured resistance agree with simulation?

Answer. *[Answer here.]*

> **Question 11.** What physical effects could account for differences?

Answer. *[Answer here.]*

> **Question 12.** Can the measured input resistance alone establish the antenna's radiation efficiency? Explain.

Answer. *[Answer here.]*

Discussion

[Resonant length vs. $\lambda/2$; what sets bandwidth; tape vs. round wire; feed-cable effects; VNA accuracy and de-embedding; what theory predicts reliably vs. what needs simulation.]